



Indian Institute of Information Technology Vadodara

Someshwar Sahu

Education

Indian Institute of Information Technology Vadodara

B.Tech in Computer Science and Engineering **CPI: 7.21/10.0**

Aug 2024 – May 2028

Gandhinagar, Gujarat, India

Suryabali Singh Public School

Class XII (Higher Secondary) – CBSE **Percentage: 88.80%**

Apr 2023 – Mar 2024

Jaunpur, Uttar Pradesh

St. Patrick's Sr. Sec. School

Class X (Secondary) – CBSE **Percentage: 87.00%**

Apr 2021 – Mar 2022

Jaunpur, Uttar Pradesh

Projects

StreamFlix – Full-Stack Streaming Platform | *React, React Native, Electron, FastAPI, PostgreSQL* | GitHub

- Solo-built a Netflix-style streaming platform spanning Web (React/Vite), Mobile (React Native), and Desktop (Electron) apps on a single FastAPI backend, exploring end-to-end video streaming architecture.
- Implemented adaptive multi-resolution HLS playback (1080p/720p/480p) with automatic and manual quality switching, powered by asynchronous transcoding via Celery workers, Redis queues, and FFmpeg.
- Designed multi-profile user accounts with a two-step auth flow (account token + profile token), cross-device watch-progress sync, watchlists, and an admin dashboard for content, user, and storage management.
- Established a shared monorepo (pnpm workspaces) with a common API client and TypeScript type package across all three clients, maintained via versioned architecture and phase-tracking documentation.

Jewellery Billing App – GST Billing & Business Management | *Python, FastAPI, SQLAlchemy, PostgreSQL* | GitHub

- Engineered a GST-compliant billing system deployed live at a jewellery shop for 30+ days, handling sale/purchase invoicing with automatic CGST/SGST/IGST calculation across 5 payment modes.
- Architected a double-sided party ledger unifying invoices, credit/debit notes, advances, and opening balances into a single live net balance per party, with oldest-due-first partial settlement.
- Integrated Google Gemini Vision to extract fields from photographed handwritten bills, and generated GSTR-1/GSTR-3B tax reports with month-level locking and formatted .xlsx exports.
- Tested by multiple jewellery shop owners in a live GST billing context, surfacing and fixing 20+ bugs across invoicing, ledger, and tax calculation flows.

Achievements

- Solved 300+ problems across LeetCode, CodeChef, and Codeforces, with a peak contest rating of **1521** on LeetCode.
- Actively competes on CodeChef and Codeforces with regular contest participation, reflecting sustained algorithmic problem-solving practice across multiple platforms.

Technical Skills

- **Languages:** C, C++, Python, C#, SQL, HTML, CSS, JavaScript, TypeScript
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems
- **Frameworks & Libraries:** React.js, React Native, Electron, Flask, FastAPI, Tailwind CSS, Node.js
- **Tools & Technologies:** PostgreSQL, SQLite, Firebase, Supabase, Celery, Redis, FFmpeg, Docker, Git, GitHub, VS Code